

Mathematical Framework and Data Harmonisation for a Macroprudential Early-Warning System

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Abstract

This companion note documents the mathematical framework, cross-country data harmonisation, and reproducibility architecture underlying the study *Early Warning Systems for Systemic Banking Distress: Persistence, Covariates, and Predictive Performance in Small Macroprudential Panels*. It formalises the construction of the harmonised banking indicators and distress target, the timing of lagged information, and the recursive out-of-sample forecasting models. Particular attention is given to the distinction between publicly available country-specific source information, study-defined harmonised measures, and the analysis-ready panel used by the forecasting code. The note thereby provides a transparent mathematical and data-construction record linking the underlying source information to the empirical early-warning system.

1 Panel notation

Let $i = 1, \dots, N$ index countries and t index years. The empirical panel covers Botswana, Malawi, Namibia, South Africa, Tanzania and Zambia. For each country-year, the principal harmonised banking indicators are equity-to-assets ratio (EAR), return on assets (ROA), loan-loss ratio (LLR), liquidity ratio (LIQ), and wholesale-funding share (WFS).

The broader financial-information vector may be written as

$$X_{it} = (EAR_{it}, ROA_{it}, LLR_{it}, LIQ_{it}, AG_{it}, WFS_{it}, CDL_{it})^\top,$$

where AG and CDL denote additional financial variables used in broader specifications. The baseline discriminative forecasting model uses the four lagged predictors

$$X_{i,t-1}^{(B)} = \begin{pmatrix} EAR_{i,t-1} \\ ROA_{i,t-1} \\ LLR_{i,t-1} \\ LIQ_{i,t-1} \end{pmatrix}.$$

WFS contributes to construction of the liquidity-fragility component of the outcome but is not a baseline continuous lagged predictor.

2 Data harmonisation and indicator construction

2.1 Harmonisation principle

Underlying banking-sector financial information was manually compiled from publicly available publications of the respective national central banks, with assistance from data collectors.

Instead, economically corresponding underlying financial items were mapped to common banking constructs and study-defined formulae were applied to construct harmonised indicators. Thus, country-specific source items $\longrightarrow$ common economic constructs $\longrightarrow (EAR, ROA, LLR, LIQ, WFS)$.

The objective is to improve definitional consistency and cross-country comparability. Harmonisation is therefore at the level of the economic construct: source-field names and, where necessary, the exact accounting components can differ across countries while representing the same intended construct.

2.2 Equity-to-assets ratio

Let TE_{it} denote total equity and TA_{it} total assets. The equity-to-assets ratio is

$$EAR_{it} = \frac{TE_{it}}{TA_{it}}. \quad (1)$$

This definition is common to all six country workbooks. A zero or unavailable denominator produces a missing value rather than an artificial numerical ratio.

2.3 Return on assets

Let P_{it} denote profit for the year. Average assets are

$$\overline{TA}_{it} = \frac{TA_{it} + TA_{i,t-1}}{2}, \quad (2)$$

and the harmonised profitability measure is

$$ROA_{it} = \frac{P_{it}}{\overline{TA}_{it}} = \frac{P_{it}}{\frac{1}{2}(TA_{it} + TA_{i,t-1})}. \quad (3)$$

The workbook formulae implement this average-assets construction subject to the availability of a valid denominator and prior asset observation.

2.4 Loan-loss ratio

Let IMP_{it} denote the applicable loan-impairment measure and $LOAN_{it}$ the corresponding loan or advance exposure. The common conceptual definition is

$$LLR_{it} = \frac{IMP_{it}}{LOAN_{it}}. \quad (4)$$

For Malawi, South Africa, Botswana and Tanzania, the principal workbook mapping is

$$IMP_{it} = \text{Loan Impairment Charges}_{it}, \quad LOAN_{it} = \text{Loans and Advances to Customers}_{it}.$$

For Namibia,

$$IMP_{it} = \text{Loan Impairment Charges}_{it}, \quad LOAN_{it} = \text{Other Loans and Advances}_{it}.$$

Zambia uses the source-specific mapping

$$LOAN_{it}^{ZM} = \text{Credit to Other Institutions}_{it} + \text{Credit to Government}_{it}, \quad (5)$$

$$IMP_{it}^{ZM} = \max\{0, -\text{Net Reversal of Impairment}_{it}\}, \quad (6)$$

so that

$$LLR_{it}^{ZM} = \frac{\max\{0, -\text{Net Reversal of Impairment}_{it}\}}{\text{Credit to Other Institutions}_{it} + \text{Credit to Government}_{it}} \quad (7)$$

when the denominator is non-zero. This country-specific mapping preserves the intended impairment-to-credit-exposure construct without assuming identical source-account labels.

2.5 Liquidity ratio

Let LA_{it} denote the harmonised stock of narrow liquid assets. The study-defined liquidity ratio is

$$LIQ_{it} = \frac{LA_{it}}{TA_{it}}. \quad (8)$$

For Malawi, South Africa and Tanzania, the workbook constructs liquid assets from the available cash/balance, cash-equivalent and interbank-placement fields:

$$LA_{it} = CB_{it} + CCE_{it} + POB_{it}, \quad (9)$$

where CB denotes cash and balances with the central bank, CCE cash and cash equivalents, and POB placements with other banks. Botswana follows the same economic construction using its corresponding field labels. Namibia uses cash and balances with banks, cash and cash equivalents, and placements with other banks.

For Zambia, the available source structure gives

$$LA_{it}^{ZM} = \text{Cash and Bank Accounts}_{it} + \text{Inventories of Domestic Cash}_{it}, \quad (10)$$

and hence

$$LIQ_{it}^{ZM} = \frac{\text{Cash and Bank Accounts}_{it} + \text{Inventories of Domestic Cash}_{it}}{TA_{it}}. \quad (11)$$

The resulting measure is therefore a harmonised narrow-liquid-assets-to-total-assets construct rather than a mechanical reproduction of a jurisdiction-specific regulatory liquidity ratio.

2.6 Wholesale-funding share

Let WF_{it} denote identified wholesale or non-customer funding and TL_{it} total liabilities. The common construct is

$$WFS_{it} = \frac{WF_{it}}{TL_{it}}. \quad (12)$$

For Botswana, South Africa and Tanzania, the workbook uses the general structure

$$WF_{it} = DB_{it} + STB_{it} + LTB_{it} + DS_{it}, \quad (13)$$

where DB denotes deposits from banks, STB short-term borrowings, LTB long-term borrowings/bonds issued, and DS debt securities or other borrowed funds. Namibia uses the corresponding debt-instruments-issued field for the final component.

For Malawi, the available mapping is

$$WF_{it}^{MW} = \text{Deposits from Banks}_{it} + \text{Long-Term Borrowings/Bonds Issued}_{it} \quad (14)$$

$$+ \text{Debt Securities and Other Borrowed Funds}_{it}. \quad (15)$$

For Zambia,

$$WF_{it}^{ZM} = \text{Credit from Financial Institutions}_{it} + \text{Credit from Other Institutions}_{it}, \quad (16)$$

and therefore

$$WF_{it}^{ZM} = \frac{\text{Credit from Financial Institutions}_{it} + \text{Credit from Other Institutions}_{it}}{TL_{it}}. \quad (17)$$

2.7 Country-level mapping summary

Table 1: Study-defined harmonised indicators and country-specific source mappings

Indicator	Countries	Workbook construction / mapping
EAR	All six	Total Equity / Total Assets.
ROA	All six	Profit for the Year / average of current and previous Total Assets, subject to denominator/prior-observation availability.
LLR	Malawi, South Africa, Botswana, Tanzania	Loan Impairment Charges / Loans and Advances to Customers.
LLR	Namibia	Loan Impairment Charges / Other Loans and Advances.
LLR	Zambia	Positive impairment-charge proxy, $\max(0, -\text{Net Reversal of Impairment})$, divided by Credit to Other Institutions + Credit to Government.
LIQ	Malawi, South Africa, Tanzania	Cash and Balances with Central Bank + Cash and Cash Equivalents + Placements with Other Banks, divided by Total Assets.
LIQ	Botswana	Corresponding cash/balance, cash-equivalent and other-bank-placement fields divided by Total Assets.
LIQ	Namibia	Cash and Balances with Banks + Cash and Cash Equivalents + Placements with Other Banks, divided by Total Assets.
LIQ	Zambia	Cash and Bank Accounts + Inventories of Domestic Cash, divided by Total Assets.
WFS	Botswana, South Africa, Tanzania	Deposits from Banks + Short-Term Borrowings + Long-Term Borrowings/Bonds Issued + Debt Securities/Other Borrowed Funds, divided by Total Liabilities.
WFS	Namibia	Same conceptual structure, with Debt Instruments Issued as the debt-instrument component.
WFS	Malawi	Deposits from Banks + Long-Term Borrowings/Bonds Issued + Debt Securities/Other Borrowed Funds, divided by Total Liabilities.

Indicator	Countries	Workbook construction / mapping
WFS	Zambia	Credit from Financial Institutions + Credit from Other Institutions, divided by Total Liabilities.

2.8 Missingness and numerical zeros

A numerical zero is treated as an observed value and is distinct from a missing observation. Where a required denominator is zero or unavailable, the spreadsheet construction returns a missing value rather than imposing an artificial ratio. The analysis-ready panel preserves this distinction.

3 Fragility indicators and distress outcome

Upstream outcome construction. The following equations formalise the construction of the composite stress score and forecasting target. In the archived implementation, these variables are supplied in the analysis-ready panel; they are not reconstructed from the original source information by the forecasting code.

Define four binary fragility indicators:

$$D_{it}^{CAP} = \mathbf{1}\{EAR_{it} < 0.04\}, \quad (18)$$

$$D_{it}^{PROF} = \mathbf{1}\{ROA_{it} \leq 0\}, \quad (19)$$

$$D_{it}^{AQ} = \mathbf{1}\{LLR_{it} > 0.03\}, \quad (20)$$

$$D_{it}^{LIQ} = \mathbf{1}\{LIQ_{it} < 0.20 \text{ or } WFS_{it} > 0.20\}. \quad (21)$$

A missing component does not independently trigger a fragility flag. For the liquidity component, an available LIQ or WFS observation can trigger the flag when its respective threshold is breached.

The composite stress score is

$$S_{it} = D_{it}^{CAP} + D_{it}^{PROF} + D_{it}^{AQ} + D_{it}^{LIQ}. \quad (22)$$

The baseline severe-distress target is

$$y_{it} = \mathbf{1}\{S_{it} \geq 2\}. \quad (23)$$

The baseline mathematical framework therefore treats $S_{it} \geq 2$ as the principal forecasting target. Alternative thresholds and onset-focused specifications belong to robustness analysis rather than the baseline target definition.

4 Lag engineering and information timing

To avoid contemporaneous information leakage, the baseline continuous predictors are lagged one year within each country:

$$EAR_{it}^{(-1)} = EAR_{i,t-1}, \quad (24)$$

$$ROA_{it}^{(-1)} = ROA_{i,t-1}, \quad (25)$$

$$LLR_{it}^{(-1)} = LLR_{i,t-1}, \quad (26)$$

$$LIQ_{it}^{(-1)} = LIQ_{i,t-1}. \quad (27)$$

Equivalently,

$$\boxed{X_{i,t-1}^{(B)} = L_i\{X_{it}^{(B)}\}}, \quad (28)$$

where L_i is the lag operator applied separately within country i . Consequently, the final observation of one country cannot become the lag of the first observation of another country.

The previous outcome used by persistence benchmarks is likewise

$$y_{it}^{(-1)} = y_{i,t-1}. \quad (29)$$

5 Recursive out-of-sample forecasting design

Let $k_0 = 3$ denote the initial country-specific warm-up period. For a forecast at year t , model estimation uses only observations dated strictly before t :

$$\mathcal{T}_t = \{(i, s) : s < t\}. \quad (30)$$

The training window expands recursively as t increases. The baseline released implementation therefore generates genuine recursive forecasts conditional on the historical information available before the forecast year.

6 Benchmark forecasts

6.1 Historical-prevalence benchmark

Let

$$\bar{y}_t^{pool} = \frac{1}{|\mathcal{T}_t|} \sum_{(j,s) \in \mathcal{T}_t} y_{js} \quad (31)$$

denote pooled historical distress prevalence. The naive benchmark is

$$\boxed{p_{it}^{NAIVE} = \bar{y}_t^{pool}}. \quad (32)$$

6.2 Lag-only benchmark

When the country's previous target is available,

$$\boxed{p_{it}^{LAG} = y_{i,t-1}}, \quad (33)$$

with numerical clipping used in implementation to avoid exact probabilities of zero or one in log-loss calculations. If a country-specific lag is unavailable, the historical prevalence provides the fallback.

7 Structural persistence model

Let country i have historical target observations $y_{i,s_1}, \dots, y_{i,s_n}$ before forecast year t . The exponentially weighted persistence state is initialized by

$$m_{i,1} = y_{i,s_1} \quad (34)$$

and recursively updated as

$$m_{i,r} = \alpha y_{i,s_r} + (1 - \alpha)m_{i,r-1}, \quad r = 2, \dots, n. \quad (35)$$

The baseline uses $\alpha = 0.50$.

To stabilize country-specific persistence estimates in the small panel, define

$$\lambda_{it} = \frac{n_{it}}{n_{it} + \kappa}, \quad (36)$$

where n_{it} is the number of country-specific historical observations before t and $\kappa = 2$ in the baseline. The structural probability is

$$p_{it}^S = \lambda_{it} m_{it}^- + (1 - \lambda_{it}) \bar{y}_t^{pool}, \quad (37)$$

where m_{it}^- is the latest recursively available persistence state. Probabilities are clipped to $[\varepsilon, 1 - \varepsilon]$ with $\varepsilon = 10^{-6}$ for numerical stability.

8 Regularised logistic model

Let

$$Z_{i,t-1} = \text{standardise}_{\mathcal{T}_t} \left(X_{i,t-1}^{(B)} \right) \quad (38)$$

denote predictors standardised using training-period information only. The discriminative probability is

$$p_{it}^D = \frac{1}{1 + \exp[-(\beta_0 + \boldsymbol{\beta}^\top Z_{i,t-1})]}. \quad (39)$$

Conceptually, the L2-regularised estimator can be represented by the penalised binary cross-entropy objective

$$(\hat{\beta}_0, \hat{\boldsymbol{\beta}}) = \arg \min_{\beta_0, \boldsymbol{\beta}} \left\{ - \sum_{(i,s) \in \mathcal{T}_t} [y_{is} \log p_{is}^D + (1 - y_{is}) \log(1 - p_{is}^D)] + \lambda \|\boldsymbol{\beta}\|_2^2 \right\}. \quad (40)$$

Here $\lambda \geq 0$ controls the strength of L2 regularisation, which shrinks the slope coefficients toward zero and stabilizes estimation in the small-sample setting. The intercept β_0 is not included in the penalty term.

The released implementation uses L2-regularised logistic regression with the **lbfgs** solver and $C = 1$. Training observations lacking the target or any required baseline predictor are excluded before model fitting. The remaining training predictors are standardised using parameters estimated from the recursively available training sample. At forecast time, missing predictor values are median-imputed using the corresponding medians learned from that training sample. When fewer than eight complete training observations are available, or when the training target contains only one class, the discriminative forecast defaults to the recursively available pooled historical prevalence.

9 Hybrid model

Define

$$\ell(p) = \log \left(\frac{p}{1-p} \right). \quad (41)$$

The hybrid combines structural and discriminative information on the log-odds scale:

$$\ell(p_{it}^H) = a + (1-w)\ell(p_{it}^S) + w\ell(p_{it}^D), \quad 0 \leq w \leq 1. \quad (42)$$

Equivalently,

$$\ell(p_{it}^H) = \ell(p_{it}^S) + a + w\{\ell(p_{it}^D) - \ell(p_{it}^S)\}. \quad (43)$$

Hence

$$p_{it}^H = \frac{1}{1 + \exp[-z_{it}]}, \quad z_{it} = a + (1-w)\ell(p_{it}^S) + w\ell(p_{it}^D). \quad (44)$$

In the released implementation, the offset-binomial GLM is estimated with a small L2 stabilisation term ($\eta = 10^{-4}$); if regularised fitting fails, an unregularised GLM fit is used. The resulting raw pooling coefficient is then clipped to $[0, 1]$. Where the hybrid estimation sample is insufficient or contains only one outcome class, the implementation falls back to an equal-weight average of the structural and discriminative probabilities; the same fallback is applied within a leave-one-country-out fold when its estimation sample is insufficient.

Hybrid predictions used for evaluation are obtained by leave-one-country-out cross-fitting: parameters applied to a held-out country are estimated from the other countries' recursive out-of-sample prediction rows.

10 Evaluation metrics

For out-of-sample observations $j = 1, \dots, n$ with outcomes y_j and predicted probabilities p_j , the Brier score is

$$BS = \frac{1}{n} \sum_{j=1}^n (p_j - y_j)^2, \quad (45)$$

and the root mean squared probability error is

$$RMSE = \sqrt{BS}. \quad (46)$$

Binary log loss is

$$LL = -\frac{1}{n} \sum_{j=1}^n [y_j \log p_j + (1 - y_j) \log(1 - p_j)]. \quad (47)$$

Discrimination is evaluated using the area under the receiver operating characteristic curve (AUC). Let p^+ denote the predicted probability for a randomly selected distress observation and p^- the predicted probability for a randomly selected non-distress observation. Then AUC admits the ranking interpretation

$$\text{AUC} = \Pr(p^+ > p^-) + \frac{1}{2} \Pr(p^+ = p^-). \quad (48)$$

Thus, AUC measures the probability that the forecasting model assigns a higher risk score to a randomly selected distress observation than to a randomly selected non-distress observation, with half credit assigned to ties.

Average precision (AP) is additionally reported to summarise performance across the precision–recall curve, particularly where the distress outcome is imbalanced.

11 Robustness specifications

Robustness analysis is conceptually separate from the baseline target. The early-fragility and strict targets are

$$y_{it}^{early} = \mathbf{1}\{S_{it} \geq 1\}, \quad (49)$$

$$y_{it}^{strict} = \mathbf{1}\{S_{it} \geq 3\}. \quad (50)$$

For the onset-focused (purified) environment, define

$$C_{it} = \mathbf{1}\{y_{it} = 1 \text{ and } y_{i,t-1} = 1\}. \quad (51)$$

Rows with $C_{it} = 1$ are excluded, so the target itself remains the baseline severe-distress indicator while continuing severe-distress observations are removed from that robustness sample.

Additional robustness exercises include alternative shrinkage values $\kappa \in \{1, 2, 5\}$, leave-one-country-out influence analysis, weight-of-evidence transformations, and macro-financial proxy augmentation.

12 Data provenance and reproducibility

12.1 Provenance of the harmonised panel

The analysis-ready panel released for reproducibility is a harmonised research dataset constructed for this study and should be distinguished from the original source material published by the respective national central banks. The original institutional source files, statistical publications, annual reports, and other source materials are not redistributed as part of the replication package. Country and year are retained in the released dataset as panel identifiers. Let i denote country and t denote year, so that each observation is indexed by the country–year pair (i, t) . These identifiers define the cross-sectional and temporal structure of the panel and are required for country-specific lagging, recursive forecasting, and leave-one-country-out analyses.

The financial indicators in the released panel are study-defined harmonised measures constructed from economically corresponding source information using the definitions and country-specific mappings documented in this note. In particular, the harmonisation process may be represented as

$$\boxed{\begin{array}{l} \text{publicly available source information} \longrightarrow \text{common economic constructs} \\ \longrightarrow \text{study-defined harmonised indicators} \longrightarrow \text{analysis-ready panel} \end{array}} \quad (52)$$

Where a national central bank separately provides statistical datasets, pre-calculated prudential ratios, or similarly named indicators, those reported values were not substituted for the study-defined harmonised measures. Instead, the common definitions and construction procedures documented above were applied consistently across jurisdictions using the most closely corresponding underlying financial information available for each country.

Accordingly, the released values of *EAR*, *ROA*, *LLR*, *LIQ*, and *WFS* should not be interpreted as official harmonised series produced by the respective central banks. They are processed research data generated through the study’s cross-country harmonisation procedure. The distinction can be summarised as

$$\boxed{\underbrace{(i, t)}_{\text{panel identifiers}} + \underbrace{(EAR_{it}, ROA_{it}, LLR_{it}, LIQ_{it}, WFS_{it})}_{\text{author-constructed harmonised measures}}} \quad (53)$$

with the underlying institutional source material remaining attributable to the respective data-producing authorities.

12.2 Reproducibility boundary

The research workflow should be interpreted as

central-bank publications and public source information $\longrightarrow$ country-specific source information
 $\longrightarrow$ study-defined harmonised indicators $\longrightarrow$ analysis-ready panel
 $\longrightarrow$ fragility target and lags $\longrightarrow$ recursive EWS models.

The original country workbooks constitute the underlying audit trail for the source-information-to-harmonised-indicator transformation, while the released forecasting code begins from the analysis-ready panel and reproduces the subsequent target environments, recursive forecasts, model comparisons, and robustness exercises. Accordingly, the mathematical definitions of *EAR*, *ROA*, *LLR*, *LIQ*, and *WFS* document upstream data harmonisation and indicator construction, whereas the forecasting equations document operations performed from the harmonised panel onward.

The released replication materials therefore begin at the harmonised analysis-ready stage and provide the processed research data, methodological documentation, and forecasting code required to reproduce the empirical analysis.

12.3 Cross-country measurement consistency

The harmonisation procedure does not assert that the accounting systems of the six jurisdictions are identical. Rather, it avoids treating similarly named central-bank ratios as automatically equivalent. Common study-defined economic constructs are formed from the most closely corresponding underlying financial items available for each jurisdiction. Country-specific mappings are retained where source structures differ. This approach improves definitional consistency while making the remaining measurement differences explicit and auditable.